

ASHMIT DEB

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EDUCATION

University of Michigan

Aug. 2023 – May 2027

B.S. in Computer Science + Minor in Entrepreneurship

Ann Arbor, MI

Cumulative GPA: 3.4/4.0

Leadership: Michigan Financial and Math Society (Quant Lead) & Michigan Student AI Laboratory (AI Project Director)

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Computer Organization, Data Science and Statistical Analysis, User Interface Development, Computer Vision, Software Engineering

EXPERIENCE

Software Engineer Intern

May 2025 – Aug. 2025

Sigray

Benicia, CA

- Enhanced development processes in Python with .NET spectral-analysis modules and NUnit tests for a microscopic X-Ray boosting feature while enabling CI/CD validation via Jenkins on Azure AKS (Remote Access API + GUI)
- Integrated Machine Learning spectral decomposition by implementing k-means clustering in Jupyter, thereby embedding fully automated mineral-phase segmentation directly into the scalable data analysis pipeline
- Built a REST Node.js microservice on AWS EC2 to stream real-time scanned X-Ray data to a Vue.js frontend dashboard, cutting preview latency by 30% via telemetry tuning and doubling productivity among engineers

Software Engineer Intern

May 2024 – Aug. 2024

Blue Cross Blue Shield

Detroit, MI

- Designed a REST-based claims processing system with Java and Spring Boot integrating PostgreSQL-backed validation logic to power an Angular frontend, reducing processing times by 40% through scalable workflows
- Designed and tuned MySQL schemas in AWS RDS, adding unit-tested queries and SharePoint validation dashboards, boosting data integrity and automated schema migrations for repeatable zero downtime releases
- Expanded the iOS and Android membership app by improving the eligibility/benefits-lookup feature through fixing 15+ bugs, integrating the React Native UI through Firebase to a Java Spring Boot backend for further security

Software Engineer Intern

May 2023 – Aug. 2023

Phaseify

Remote

- Developed custom full-stack web application for the career-guidance platform implementing responsive UIs with React on the frontend and MongoDB on the backend, improving scalability and cutting manual coordination by 25%
- Built a shell scripting-based automation tool to parse JSON and mask sensitive data, expediting the setup of testing environments for the claims adjudication engine and saving 3+ hours weekly
- Converted Figma feedback into modular features to streamline workflows across custom platforms translating designs into REST APIs with schema checks, error codes, and versioned routes lowering regressions and rework

PROJECTS

Reddit API Stock Analyzer

Jan. 2025 – Feb. 2025

- Developed a multithreaded C++ app that pulls finance-subreddit posts (r/WSB, r/stocks, r/investing) via the Reddit API (cURL + JsonCpp), scores sentiment, and outputs to a live CLI and auto-generated HTML report in real time
- Applied modern C++ practices (RAII, STL) and thread-pool concurrency with mutex-guarded queues, plus lock-free metrics, to accelerate data ingestion and scoring, producing near-real-time, actionable stock-sentiment insights

Real-Time Earthquake Monitor

Sep. 2024 – Nov. 2024

- Built a Node.js and Express.js service using USGS GeoJSON and streaming live updates via Socket.io and storing data on PostgreSQL while applying Flask microservice for real-time statistical analysis through RESTful API
- Used HTML, CSS, JavaScript, and React on frontend and deployed PostgreSQL, Node API, and Flask analytics with Docker Compose pushing images to AWS ECR and securely scaling on ECS for continuous seismic tracking

SKILLS

Languages: C#, C++, C, Swift, Python, Java, SQL, JavaScript, Go, HTML, CSS, Kotlin, Perl, PHP

Frameworks/Libraries: Node.js, Vue.js, React.js, Angular, .NET, Selenium, Jupyter, Firebase, NoSQL

Cloud/Tools: Azure (ACR, AKS), AWS (RDS, EC2, ECS), Figma, Git, CI/CD, Docker, Jenkins, Jira, Grafana